
OenoBench: A Wine-Domain Benchmark for Knowledge-Grounded Evaluation of Large Language Models

Anonymous Author(s)

Affiliation

Address

email

Abstract

We introduce **OenoBench**, a knowledge benchmark for the wine domain comprising **3,266 multiple-choice questions** across six domain pillars (regions, grape varieties, viticulture, winemaking, producers, business) and four difficulty tiers. The benchmark is built from **38,104 atomic, source-anchored facts** extracted by 35 provenance-verified scrapers from authoritative sources spanning government registries (INAO, TTB, OIV), peer-reviewed journals, university research groups, and Wikipedia/Wikidata. Our central methodological contribution is a *LLM-driven, end-to-end pipeline* in which large language models reformat verified facts into questions and audit the result, but never serve as the source of truth: every claim traces to a URL with a tier-of-authority label, every question is generated by one of five complementary strategies (fact-to-question, template, comparative, scenario, distractor mining) distributed across five generator models (Claude, GPT, Gemini, Llama, Qwen + deterministic templates), and every question is scored by a nine-agent automated audit organised into four teams and calibrated against a human gold sheet via Cohen’s κ . We evaluate sixteen frontier configurations and report five findings: (i) overall accuracy ranges from 53% to 84%, with reasoning-mode **o3** on top at **83.6%**; (ii) reasoning-mode lift is concentrated in **DeepSeek R1 (+6.8 pp)** and largely absent for Claude Opus and Gemini Pro; (iii) the Anthropic family shows a **+9 pp self-preference** effect on its own questions while Google models show a **8 pp inverse** effect; (iv) frontier open-weight models occupy the cost-vs-accuracy Pareto frontier together with proprietary reasoning models; and (v) every config gains **≈ 33 pp** on questions that the audit’s closed-book panel solves from world knowledge alone, revealing a parametric-recall ceiling that only the contextual, source-grounded slice avoids. We release the corpus, the audit findings, the human-review web app, and the construction code under CC-BY-SA-4.0.

1 Introduction

Large language models are increasingly evaluated on knowledge-intensive benchmarks, but the most widely-used resources — MMLU [6], GPQA [16], and their successors [19, 21, 15] — share three properties that limit their usefulness as the field matures: (i) they aggregate breadth at the expense of *depth* within any single discipline, (ii) they are typically authored or curated by a small team using a single generation pipeline, exposing them to systematic blind-spots and stylistic regularities that frontier models can learn to exploit [14, 20], and (iii) the questions are increasingly suspected of contamination through web-scale pre-training corpora [17, 13, 3]. As models grow more capable, the marginal value of a benchmark depends increasingly on *how* it was constructed, not only on what it contains.

We argue that domain-specialised benchmarks built around bodies of knowledge that are *externally validated by humans* — through certification, regulation, or peer-reviewed practice — offer a useful complement to broad knowledge benchmarks. Examples already exist in medicine [9, 10], law [5], and finance [8]. We extend this line of work to a domain that has so far received little attention from the benchmarking community, despite being unusually well-defined: **wine**.

Why wine. Wine is a multi-disciplinary subject that integrates plant biology, geology and climate science, organic chemistry, food microbiology, sensory neuroscience, regulatory law, and business economics. It is one of very few domains for which (a) authoritative governmental and inter-governmental sources publish structured taxonomic data (e.g. INAO, OIV, TTB, regional consortia), (b) university research groups (notably UC Davis, Geisenheim, Bordeaux Sciences Agro) produce open scholarly literature, and (c) professional certification bodies define a graded competence ladder (WSET Levels 1–4, Court of Master Sommeliers, Institute of Masters of Wine) whose published syllabi can be used to ground what “expert” should mean. Together these properties make it possible to build a benchmark in which every question is traceable to a verifiable source, while still spanning the recall–reasoning–application spectrum that distinguishes a strong taster, winemaker, or buyer from a memorised glossary.

Contributions.

1. **A 3,266-question wine benchmark** covering six pillars (regions, varieties, viticulture, winemaking, producers, business), built from 38,104 atomic facts whose provenance is recorded down to the source URL and tier of authority (Section 3).
2. **An LLM-driven, fact-grounded question-generation pipeline** that combines five generation strategies and five generator models. The LLM’s role is reformatting verified facts, *not* providing them; an explicit closed-book gate flags questions solvable from world knowledge alone, and a per-strategy quota prevents pipeline mono-culture.
3. **A nine-agent automated quality audit** (Section 4) consisting of static, statistical, and tri-judge agents organised into four teams, calibrated against a human gold sheet using Cohen’s κ . The audit removed 341 questions and re-labelled the difficulty of 1,259 in the final release. Crucially, the audit also revealed that one of its own agents (closed-book solvability) over-reports leakage relative to humans; we kept the affected questions with explicit disclosure rather than dropping them, and demonstrate the consequences in the evaluation.
4. **A bias-aware evaluation across sixteen frontier configurations** (Section 5) reporting overall accuracy, reasoning-mode lift, a Self-Preference Score that quantifies own-vs-other-generator gaps, a cost-efficiency Pareto analysis, and a closed-book vs. source-grounded contrast that isolates parametric memorisation from contextual reasoning.

OenoBench, the construction and audit pipelines, the multi-agent audit findings, and the human-review web application are released under CC-BY-SA-4.0; see Appendix E for dataset and code URLs and the double-blind anonymisation map.

2 Related Work

Broad knowledge benchmarks. MMLU [6], BIG-Bench [19], AGIEval [21], GPQA [16], HELM [12], and Humanity’s Last Exam [15] measure factual breadth and reasoning across many subjects. As headline scores have saturated, attention has shifted toward harder reasoning benchmarks and toward auditing the construction process. Our work is complementary: rather than maximising difficulty across many disciplines, we maximise depth within a single externally-validated knowledge domain, and treat *construction-time bias control* as a first-class methodological contribution rather than a downstream decontamination step.

Domain-specific benchmarks. MedQA [9], PubMedQA [10], LegalBench [5], and FinanceBench [8] demonstrate the value of grounding evaluation in a regulated profession with clear sources of truth. We adopt their organising principle — expert syllabus as taxonomy, source-traceable facts as backbone — and add multi-model question generation, an automated multi-agent audit, and self-preference reporting as additional bias controls. To our knowledge, OenoBench is the first such benchmark for the wine domain.

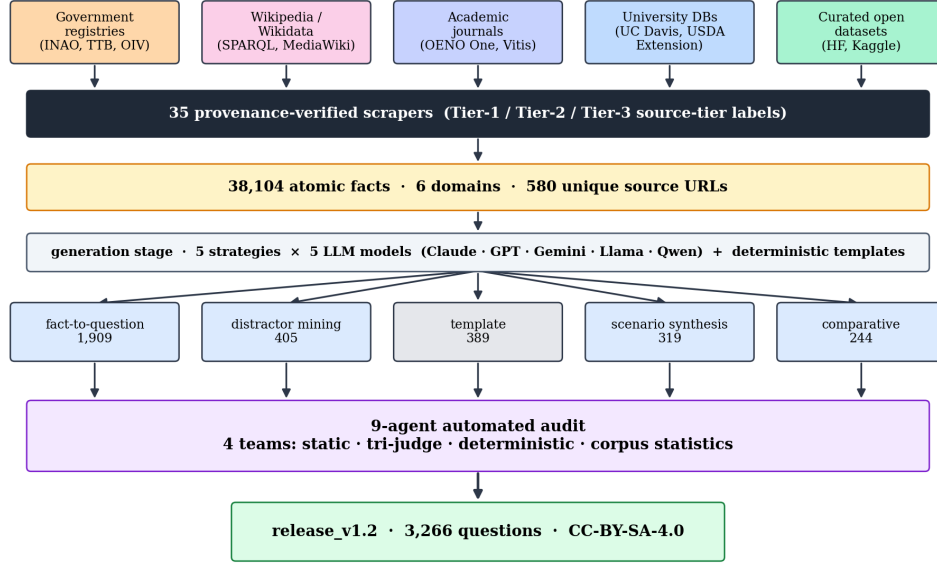


Figure 1: End-to-end pipeline. Authoritative sources (top) are scraped by 35 source-specific extractors that share an atomic-fact decomposition and validation pass. Generated questions are gated by a closed-book pre-screen and a nine-agent audit before being tagged for release.

88 **LLM-as-judge bias and self-preference.** A growing literature documents that models tend to
 89 favour outputs stylistically similar to their own [14, 20, 18]. This biases benchmarks where the
 90 same model generates and grades. We mitigate this in two ways: by distributing generation across
 91 five models (Claude, GPT, Gemini, Llama, Qwen) plus deterministic templates, and by reporting
 92 per-model self-vs-other accuracy gaps as a first-class diagnostic (Section 5.3).

93 **Benchmark contamination and solvability.** Pre-training exposure inflates apparent capability
 94 [17, 13, 3]. We mitigate contamination in two ways: (i) generating questions from atomic facts rather
 95 than copying source text (a fact-echo agent flags any question whose longest common substring with
 96 the source exceeds 65%), and (ii) running an explicit *closed-book solvability* pre-screen and audit,
 97 allowing us to report a separate per-config accuracy on world-knowledge-solvable items (Section 5.5).
 98 We further show that the same audit, when calibrated against a human gold sheet, reveals that
 99 LLM judges themselves over-report leakage by an order of magnitude — a finding with broader
 100 implications for the LLM-as-judge paradigm.

101 **Wine in NLP.** Wine has previously appeared in NLP only as a substrate for review-text generation,
 102 recommendation [2], taxonomy learning [11], and judge-reliability studies [7]. OenoBench is the first
 103 benchmark that targets *structured factual knowledge* of viticulture, winemaking, and wine regions
 104 evaluated against frontier LLMs.

105 3 Dataset Construction

106 OenoBench is constructed by an LLM-driven pipeline (Figure 1) in two stages: (1) a *fact-collection*
 107 stage that builds a corpus of atomic, provenance-tagged statements from authoritative sources,
 108 and (2) a *question-generation* stage that converts those facts into benchmark questions using five
 109 complementary strategies and five generator models. The pipeline is designed around a single
 110 methodological commitment: large language models are used as *rephrasers and auditors*, never as the
 111 source of truth. Every fact in the corpus traces to a URL with a tier-of-authority label, every question
 112 is anchored in one or more such facts, and every question is scored by a nine-agent automated audit
 113 (Section 4) before release.

Table 1: Atomic-fact corpus: 38,104 facts from 35 scrapers, six domain pillars, three source tiers.

Domain pillar	Facts	Share	Tier 1	Tier 2/3
Wine regions	18,943	49.7%	23.4%	76.6%
Producers	6,215	16.3%	9.1%	90.9%
Grape varieties	5,959	15.6%	18.8%	81.2%
Viticulture	3,635	9.5%	12.0%	88.0%
Wine business	1,985	5.2%	31.4%	68.6%
Winemaking	1,367	3.6%	16.7%	83.3%
Total	38,104	100.0%	19.6%	80.4%

3.1 Domain taxonomy

We organise wine knowledge into six *domain pillars*, chosen to align with the WSET Level 3 and Diploma syllabi: (i) **wine regions** (appellations, classifications, geography, climate, soils); (ii) **grape varieties** (ampelography, parentage, regional distribution, sensory profile); (iii) **viticulture** (vineyard practice, training systems, pests and diseases, vintage effects); (iv) **winemaking** (vinification choices, fermentation, ageing, faults); (v) **producers** (estates, négociants, co-operatives, key houses); and (vi) **wine business** (markets, regulation, distribution, economics). Each domain has a separate target share informed by syllabus weighting and source availability; the realised distribution is reported in Table 1.

3.2 Source tiering, licensing, and non-fabrication

Every fact is tagged with one of three source tiers, mirroring evidence-based-medicine practice:

- **Tier 1 (official, 19.6%)**: government registries and inter-governmental bodies (INAO, OIV, TTB, regional consortia, the UC Davis AVA database).
- **Tier 2 (authoritative, 76.6%)**: Wikipedia (34.3% of facts), Wikidata SPARQL (31.0%), wine-body and consortium websites, peer-reviewed journals (*OENO One*, *Vitis*, *Australian Journal of Grape and Wine Research*).
- **Tier 3 (reliable, 3.8%)**: curated open datasets on HuggingFace and Kaggle, secondary reference databases.

Source licensing. Every source listed above is either public-record or released under a licence that permits scraping and redistribution of derivative works. Tier-1 government data is in the public domain (TTB) or under explicit open-data licences (INAO Licence Ouverte, OIV open data, UC Davis AVA database). Tier-2 Wikipedia content is CC-BY-SA-3.0/4.0 and Wikidata is CC0; peer-reviewed journals (*OENO One*, *Vitis*, *AJEV*) are open-access under Creative Commons. Tier-3 datasets on HuggingFace and Kaggle are used under their declared licences. All scraping is rate-limited and identifies itself with the disclosed user agent `OenoBench-Research/1.0`; per-fact source URLs are released alongside the corpus so that downstream users can verify and re-attribue. The full source-by-source licence table is in Appendix E.

Non-fabrication guarantee. A central design commitment of OenoBench is that *no fact in the corpus is sourced from an LLM’s internal knowledge*. Every fact is extracted by a source-specific scraper from a live HTTP, SPARQL, or file-based source and is stamped with the source URL, retrieval timestamp, and tier label before insertion. The released codebase enforces this discipline structurally: scrapers do not have access to generative-LLM clients, and the fact-insertion utility rejects records without a verified source-URL trace. We make this guarantee prominent because LLM-generated content masquerading as scraped data is a contamination vector that conventional dataset documentation does not detect, and recovering from it after-the-fact is mechanical only if the scraper code is released alongside the data.

3.3 Atomic fact extraction

Source pages are decomposed into atomic facts by a five-stage pipeline (Figure 2): (1) sentence splitting at conjunctions and clause boundaries, (2) reference resolution (pronouns and demonstratives

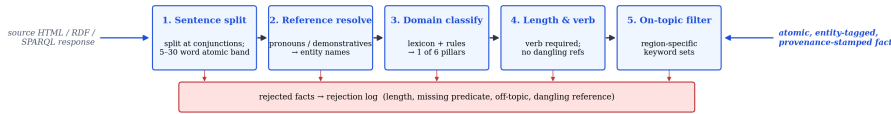


Figure 2: Atomic-fact extraction pipeline. Inputs are heterogeneous (HTML pages, RDF graphs, SPARQL responses, CSV dumps); outputs are uniform single-assertion sentences with entity tags, domain label, source URL and tier.

are replaced with their entity referents from article context), (3) domain classification into one of the six pillars, (4) a length-and-predicate validator that rejects facts shorter than five words, longer than fifty words, missing a verb, or carrying dangling references, and (5) an on-topic filter parameterised by region-specific keyword sets to prevent cross-contamination (e.g. Austrian content appearing in a Bordeaux scraper). Wikidata SPARQL queries use direct country relations (P17) rather than transitive administrative parents (P131*); the latter caused severe cross-region leakage in early scrapers and is documented as a cautionary case in Appendix B.

Facts are required to be (a) *atomic*, one assertion per sentence; (b) *entity-tagged*, with subject and object linked to canonical IDs in the wine knowledge graph; and (c) *source-faithful*, a paraphrase of the original, never a verbatim copy. This last constraint is enforced both at extraction time and at audit time (Section 4, agent A3).

3.4 Question generation: five strategies, five models

A single-strategy, single-model benchmark inherits the blind-spots of its pipeline. We mitigate this by combining *five strategies* and *five generator models*, with per-strategy and per-model quotas enforced by an orchestrator that resumes safely across runs. Strategies are:

- **Fact-to-question (58.4%, 1,909 questions):** the LLM rewrites a single verified fact as a multiple-choice question; preserves grounding but tends toward recall.
- **Distractor mining (12.4%, 405):** confusable-entity sampling produces wrong answers that share the right answer’s category and dimension, raising plausibility and forcing fine-grained discrimination.
- **Template (11.9%, 389):** forty-five deterministic parameterised templates across the six pillars; pure entity substitution, zero LLM creativity — a baseline against which neural generation is measured.
- **Scenario synthesis (9.8%, 319):** coherent fact clusters are converted into applied multi-fact reasoning prompts (e.g. a sommelier service decision, a winemaker blending decision, a viticulturist vintage call). Domain-specific scenario types prevent persona-content mismatch.
- **Comparative (7.5%, 244):** entity-affinity scoring pairs related entities (e.g. two Burgundy premiers crus, two Rhône grapes) and asks “which differs in X” or “what do both share”.

The five generator models are Claude Opus 4.7, GPT-5, Gemini 2.5 Pro, Llama 3.1 405B, and Qwen 3.5 235B, each accessed through a unified OpenRouter client with consistent temperature and top-*p*. The realised generator distribution across the released corpus is **Qwen 20.4%, Llama 19.3%, Claude 19.0%, ChatGPT 16.6%, Gemini 12.9%, deterministic templates 11.9%** (computed directly from the 3,266 released questions). The orchestrator allocates generation share dynamically based on per-pilot audit pass rates — generators whose questions pass audit reliably get a larger share in subsequent rounds — subject to a hard **per-generator cap of 21%** to keep any single model from dominating the corpus.

Closed-book gate. Questions targeting difficulty levels 1 and 2 are passed through a *closed-book pre-screen* before admission: a separate LLM (Claude Sonnet for L1, Opus for L2) attempts the question with no source fact in context, and questions it answers correctly are tagged `closed_book_solvable` and reserved into a quota. Early pilots used a 25% cap; we raised the cap to **50%** after the gold-sheet calibration revealed that LLM judges over-attribute closed-book solvability relative to humans (Section 4.2), so the higher cap keeps the items that are in fact discriminating for downstream

evaluation. Difficulty levels 3 and 4 skip the gate by protocol, since high-difficulty items are expected to be answerable only with the supporting fact. The audit’s B2 agent (Section 4) re-runs this check at corpus scale on a panel of three judges; we discuss its calibration against humans and the implications for the eval contrast in Section 5.5.

3.5 Final corpus: release_v1.2

After generation, audit, and post-eval review, the released corpus release_v1.2 contains **3,266 questions**. The distribution along the domain and difficulty axes is:

Table 2: Composition of the released corpus.

Domain	N	%	Difficulty	N	%
Wine regions	1,108	33.9%	L1 (entry)	694	21.2%
Grape varieties	766	23.5%	L2 (intermediate)	894	27.4%
Producers	515	15.8%	L3 (advanced)	678	20.8%
Viticulture	502	15.4%	L4 (expert)	1,001	30.6%
Wine business	250	7.7%			
Winemaking	187	5.7%			
Total	3,266	100%	Total	3,266	100%

The corpus underwent three post-generation passes: a 9-agent automated audit (Section 4) that dropped 341 questions and re-labelled the difficulty of 1,259; a stratified human gold-sheet review of 50 questions per pilot release (rated by a WSET-Diploma reviewer through the released human-review web app, Appendix D, used both to calibrate audit-agent κ and to surface manual quality flags); and a post-evaluation review of 97 items that were answered incorrectly by all 16 evaluation configurations (Section 5, Appendix C), of which 54 were removed as defects, 9 were dropped after a borderline-review pass, and the remainder retained as legitimately hard items. We document the dataset following Datasheets for Datasets [4]; the full datasheet is in Appendix A and a Croissant manifest [1] ships with the HuggingFace release.

4 Multi-Agent Quality Audit

The audit pipeline runs every generated question through nine automated agents organised into four teams. Each agent emits a per-question signal in $\{\text{PASS}, \text{WARN}, \text{FAIL}\}$ and signals are calibrated against a human-reviewed gold sheet using Cohen’s κ . Agents whose human agreement falls below threshold are downweighted to advisory-only, on the principle that an automated judge whose verdict humans cannot reproduce should not gate question admission. A separate human-review web application (Appendix D) collects the gold-sheet ratings from WSET-Diploma reviewers and is itself part of the released artifact set. A team-by-team architecture diagram (Figure 8, Appendix C) shows how candidate questions flow through the four teams to the drop / relabel / keep verdict.

4.1 Team architecture

Of the 14 agents, **nine are always-run** on every candidate question; the remaining **five are escalation-gated** and are invoked only when an upstream agent flags a corpus-level concern. This deferred-execution policy keeps routine audit cost bounded ($\sim$ \$76 for the full corpus audit reported here, 5h 22m wall) while preserving full coverage when a defect class is suspected. The full audit code, prompts, and thresholds are released alongside the corpus (Appendix B).

4.2 Gold-sheet calibration

A human gold sheet of 50 stratified questions per release is rated by a WSET-Diploma reviewer through the released human-review web app (Appendix D). **15+ gold-sheet rounds** have been completed across preliminary releases (audit pilots v1–v16) so that the agent thresholds and κ values reported here represent calibration aggregated over an iterative cycle, not a single batch. Each round rates against eight rubrics: answer_correct, distractors_plausible, not_ambiguous, source_faithful, needs_source, no_vague_language, labels_correct,

Table 3: All 14 audit agents across four teams. Static agents use no LLM; tri-judge agents use a panel of three frontier models with majority voting. The *Run* column indicates whether the agent is invoked on every question (ALWAYS) or only when an upstream agent flags a corpus-level concern (ESC, for *escalation-gated*).

Team	Agent	Run	Function
A. Static	A1 LexicalHygiene	ALWAYS	Regex sweep for vague phrasing, marketing language, meta-questions.
	A2 BiasStats	ALWAYS	χ^2 on correct-answer position; Mann–Whitney U on correct-vs-distractor length.
	A3 FactEcho	ALWAYS	Longest-common-substring ratio between question text and source fact (FAIL at $\text{LCS} \geq 0.65$).
	A4 TemplateFingerprint	ALWAYS	POS-bigram logistic regression detector for template-induced stylistic regularities (held-out AUC 0.84).
B. Tri-judge	B1 TriJudgeAnswer	ALWAYS	Claude / GPT / Gemini panel reads question + source fact and votes on the marked-correct option.
	B2 ClosedBookSolvability	ALWAYS	Same panel, source fact <i>withheld</i> : can the question be answered from world knowledge alone?
	B3 UbiquityRisk	ALWAYS	Internationally-grown grape stem $\times$ region-class answer $\Rightarrow$ ambiguity flag.
	B4 Ambiguity	ESC	Tri-judge re-read flags questions with >1 defensible answer; invoked when B1 dissent rate elevated.
	B5 VerifierSkip	ESC	Self-consistency probe across re-prompted panels; invoked on disputed B1 verdicts.
C. Determin.	C1 DistractorDifficulty	ESC	Embedding-distance distractor analysis; invoked when A2 reports length skew.
	C2 CategoryLeak	ALWAYS	Wine-type distractor validation; flags red/white/sparkling category mismatches.
	C3 SourceSwap	ESC	Substitutes alternate source facts to test answer stability; invoked on B1/B2 disagreement.
	C4 DifficultyAudit	ALWAYS	Gemini Pro re-rates difficulty; FAIL/WARN when delta ≥ 2 from assigned.
D. Corpus stats	D1 SelfPreference	ALWAYS	Per-generator-family own-vs-other accuracy on a held-out probe; corpus-level signal.
	D2 DedupCalibration	ESC	Near-duplicate audit at cosine ≥ 0.92 ; invoked on suspicious cluster signals.
	D3 SkewAudit	ALWAYS	Country / sub-domain over-representation χ^2 .

and `verbatim_copy`. For each automated agent we compute Cohen’s κ against the corresponding human rubric on the gold sheet. **Agent signals with $\kappa < 0.6$ are downweighted to advisory-only and excluded from gating decisions.**

The most consequential calibration result concerns agent **B2 ClosedBookSolvability**: against the human reviewer, who solved only $\approx 12\%$ of the gold-sheet questions without seeing the source fact, the LLM tri-judge panel reported $\approx 83\%$ solvability, yielding $\kappa \approx 0.007$. We interpret this not as a defect of the benchmark but as a property of the *evaluator*: frontier LLM judges have absorbed enough wine knowledge during pre-training to over-attribute “world knowledge” to questions that humans find non-trivial. We therefore retain the 1,601 B2-flagged questions in the released corpus, document the finding in the datasheet, and report a closed-book vs. source-grounded contrast in Section 5.5 that turns the audit’s blind spot into a calibrated diagnostic of memorisation reliance.

4.3 Release-cycle results

The 9-agent audit on `release_v1.1` (3,670 candidate questions) produced the verdicts in Table 4. After drops, a follow-up post-evaluation pass (Section 5 and Appendix C) audited the 97 questions that all 16 evaluation configs answered incorrectly: 54 were removed as defects (wrong ground truth, all-correct options, duplicate options), 9 were dropped on borderline-review and the remaining 34 retained as legitimately hard. The final released corpus is 3,266 questions.

Table 4: Audit findings on `release_v1.1` and the resulting drop / re-label policy.

Audit signal	Triggered FAIL	Action
A1 LexicalHygiene (vague phrasing)	60	drop
A3 FactEcho (verbatim copy)	63	drop
B1 TriJudgeAnswer (wrong answer key)	47	drop
C2 CategoryLeak (wine-category leak)	9	drop
B3 UbiquityRisk (grape $\times$ region ambiguity)	183	drop
Distinct dropped (with overlap)	341	
C4 DifficultyAudit (relabel from 1,252 LLM hits + 7 human)	1,259	re-label
B2 ClosedBookSolvability (kept w/ disclosure)	1,601	disclose

The C4-driven difficulty re-label was decisive: it shifted the corpus from 14% L3+L4 items (the original generator-assigned distribution) to **51% L3+L4** (Table 5), bringing the hardest-tier counts above the entry-tier counts and substantially sharpening difficulty-stratified analyses (Section 5.1).

Table 5: Difficulty distribution before and after C4 re-label.

	Generator-assigned	Post-relabel	Δ
L1 (entry)	1,261 (38%)	694 (21%)	−567
L2 (intermediate)	1,559 (47%)	894 (27%)	−665
L3 (advanced)	218 (7%)	678 (21%)	+460
L4 (expert)	291 (9%)	1,001 (31%)	+710

4.4 Calibration philosophy

We emphasise the design choice underlying these numbers. A naive “automated audit” interpretation would have dropped all 1,601 B2-flagged questions, which would have removed nearly half the corpus and biased the remainder toward whatever idiosyncratic items the LLM panel happened not to know. By calibrating B2 against humans and treating it as an evaluator-side signal, we preserve dataset value while giving downstream users a principled way to interpret leakage. The audit’s role, on this reading, is not to be the final arbiter of quality, but to surface signals *that humans validate* as quality-relevant, and to disclose clearly the signals where humans and LLMs disagree.

5 Evaluation

We evaluate **16 frontier configurations** on the 3,266-question release in a single end-to-end run. The slate combines six within-family cost pairs (Claude Opus 4.7 vs. Claude Haiku 4.5, Gemini 2.5 Pro vs. Gemini 2.5 Flash, GPT-5 vs. GPT-5-mini, Llama 3.3 70B vs. Llama 3.1 8B, Qwen 2.5 72B vs. Qwen 2.5 7B), four reasoning-mode configurations (o3, DeepSeek R1, Gemini 2.5 Pro thinking, Claude Opus 4.7 thinking), and two additional standard models (DeepSeek V3, Mistral Large). Every config sees the same 3,266 questions, with single-letter output (A–D) at `max_tokens=5` and a five-stop fallback. We organise the evaluation around five questions, one per subsection.

5.1 Overall capability ranking

OenoBench cleanly stratifies the slate (Figure 3). The top tier is led by **o3 at 83.6%**, followed by GPT-5 (82.8%), Gemini 2.5 Pro thinking (82.6%), Gemini 2.5 Pro (81.7%), and Claude Opus 4.7 (81.0%, tied between standard and thinking modes). A clear band separates frontier ($\geq 80\%$) from mid-tier (mid-60s to high-70s: Mistral Large 69.1%, Llama 3.3 70B 67.1%, Qwen 2.5 72B 67.4%, DeepSeek V3 70.3%, Gemini 2.5 Flash 75.1%, GPT-5-mini 78.4%) and small-tier (low-50s to low-60s: Llama 3.1 8B 60.5%, Qwen 2.5 7B 57.0%, Claude Haiku 4.5 53.3%). No config performs at random; the spread between best and worst is 30 pp, comparable to the spread observed on GPQA [16] but on a domain corpus $30\times$ smaller and fully provenance-tagged. Difficulty-stratified accuracies (Appendix C, Table 9) confirm clean stratification on re-labelled difficulty: from 93.6% (L1) down to **58.7% on L4 expert items**, with frontier configs holding 68–71% on L4 and small configs at 39–45%.

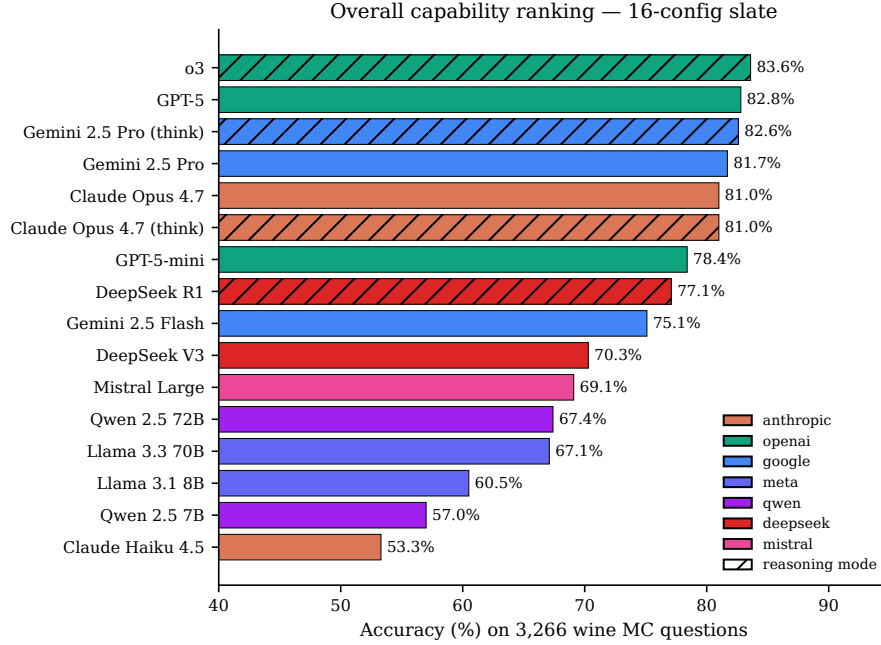


Figure 3: Overall accuracy on the 3,266-question corpus across 16 configurations. Reasoning-mode configs are hatched.

5.2 Reasoning-mode lift

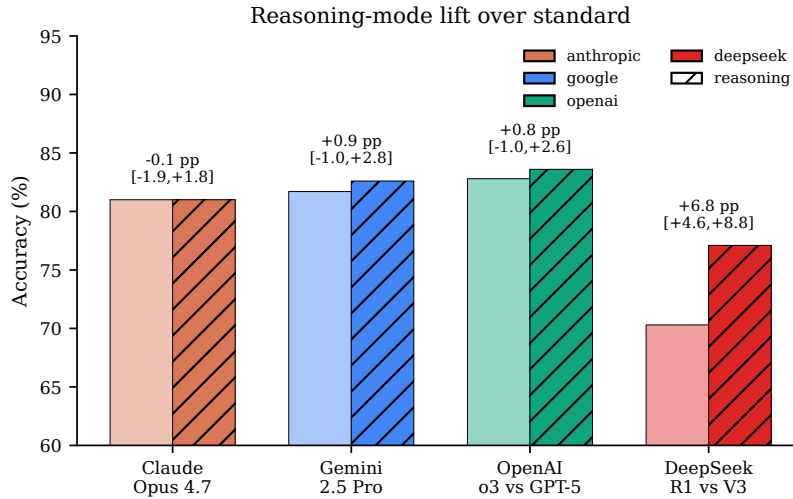


Figure 4: Accuracy of reasoning-mode vs. standard configurations for four families. Numbers above bars: δ in pp with bootstrap 95% CI from 1,000 resamples. Family colours match Figure 3; the hatched bar is the reasoning mode. Only DeepSeek shows a statistically significant lift; for the other three, zero is inside the CI.

Reasoning-mode lift is concentrated in one family (Figure 4). **DeepSeek R1 vs. DeepSeek V3:** +6.8 pp (95% CI [+4.6, +8.8]), the only paired contrast whose CI excludes zero. Each of **o3 vs. GPT-5** (+0.8 pp [-1.0, +2.6]), **Gemini 2.5 Pro thinking vs. Gemini 2.5 Pro** (+0.9 pp [-1.0, +2.8]), and **Claude Opus 4.7 thinking vs. Claude Opus 4.7** (-0.1 pp [-1.9, +1.8]) is statistically indistinguishable from zero. We attribute the DeepSeek result to DeepSeek V3 leaving substantial headroom on this domain (70.3%) which R1's chain-of-thought can recover; the three frontier

cases are likely already near the corpus ceiling on parametrically-recallable items. Per-difficulty breakdowns (Appendix C, Table 10) confirm this: reasoning helps DeepSeek across all four difficulty tiers, but the frontier reasoning configs are flat against their standard counterparts at L1–L3 and gain at most 1–2 pp at L4. The cost-vs-lift trade-off is unfavourable for reasoning configs in this domain except DeepSeek (Section 5.4).

5.3 Self-preference bias

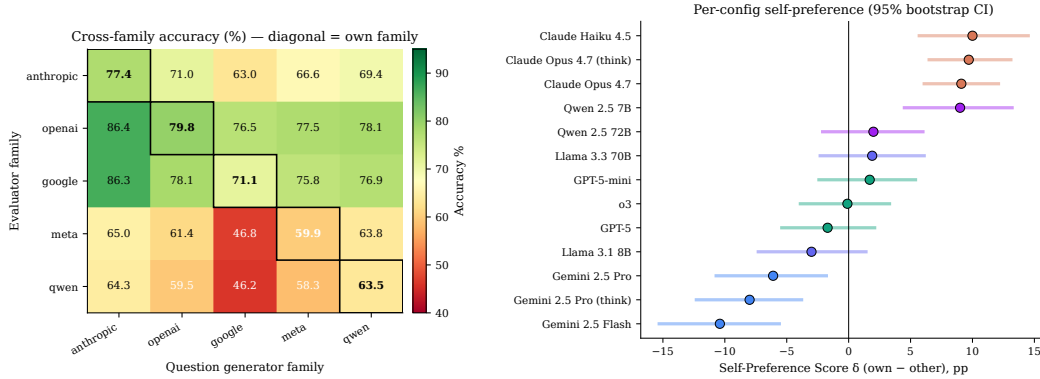


Figure 5: Self-preference bias. *Left*: cross-family accuracy matrix; rows are evaluator family, columns are question-generator family, diagonal entries are own-family. *Right*: per-config Self-Preference Score $\delta = \text{Acc}(\text{own}) - \text{Acc}(\text{other})$ with 95% bootstrap CI; family colours match Figure 3.

We compute the *Self-Preference Score (SPS)* per config m as $\text{SPS}(m) = \text{Acc}(m | Q_{\text{own}}) - \text{Acc}(m | Q_{\text{other}})$ where Q_{own} collects questions generated by m 's family and Q_{other} its complement. A pipeline that successfully neutralises generator fingerprints should produce $|\text{SPS}|$ near zero.

The result (Figure 5) is family-dependent. **Anthropic** models show a strong positive SPS: +9.1 pp [+6.1,+12.1] for Claude Opus 4.7, +9.7 pp [+6.5,+13.1] for Claude Opus 4.7 thinking, +10.0 pp [+5.7,+14.5] for Claude Haiku 4.5. **OpenAI** is statistically indistinguishable from zero across o3, GPT-5, and GPT-5-mini (−1.7 to +1.7 pp). **Google** models show an unexpected *negative* SPS: −6.1 pp [−10.7,−1.8] for Gemini 2.5 Pro, −8.0 pp [−12.3,−3.8] for Gemini 2.5 Pro thinking, and −10.4 pp [−15.3,−5.6] for Gemini 2.5 Flash. The cross-family matrix (Figure 5, left) traces the cause: Google-generated questions are uniformly harder for everyone (46–76% across rows), while Anthropic-generated questions are uniformly easier for Anthropic (77.4%) but typical for others (65–86%). We interpret the Anthropic +9 pp gap as a residual stylistic fingerprint that pre-release paraphrase passes did not neutralise; the Google −8 pp gap suggests Google generators produce questions that are harder *by phrasing*, not by leak. Both findings are first-class disclosures with the released corpus.

5.4 Cost efficiency

The 16-config slate spans nearly four orders of magnitude in evaluation cost: Llama 3.1 8B costs \$0.01 for the full run while Gemini 2.5 Pro costs \$29.47 (Figure 6). The Pareto frontier consists of five configs: **Llama 3.1 8B** (60.5% / \$0.01), **Gemini 2.5 Flash** (75.1% / \$0.12), **GPT-5-mini** (78.4% / \$2.82), **Claude Opus 4.7** (81.0% / \$3.35), and **o3** (83.6% / \$11.80). Two of the most expensive configs (Gemini 2.5 Pro at \$29.47 and GPT-5 at \$21.90) are dominated by cheaper Pareto neighbours — Gemini 2.5 Pro by o3, and GPT-5 by Claude Opus 4.7 / o3 / GPT-5-mini — because their additional spend buys output tokens, not better wine answers. Reasoning-mode lift (Section 5.2) translates into a Pareto win only for DeepSeek R1, which is itself dominated by GPT-5-mini in this corpus. The overall efficient-accuracy boundary therefore favours *small proprietary* (GPT-5-mini, Claude Opus 4.7) and one reasoning model (o3) over either reasoning-everywhere or scale-everywhere strategies.

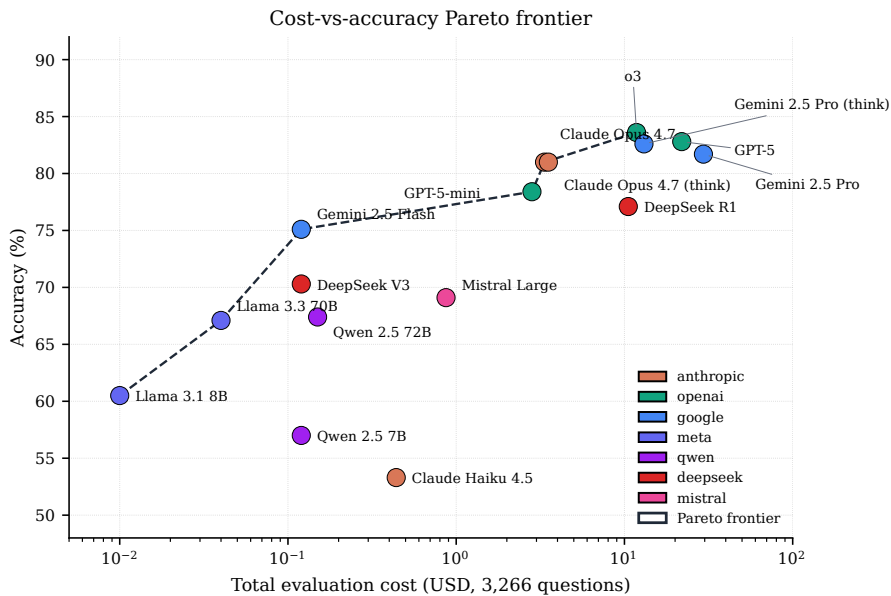


Figure 6: Total OpenRouter cost (log scale) versus accuracy on the full 3,266-question slate. Family colours match Figure 3; the Pareto frontier (red dashed) illustrates the cost-vs-quality trade-off.

5.5 Closed-book vs. source-grounded performance

Section 4.2 described the calibration finding that the audit’s B2 ClosedBookSolvability agent reports leakage at $\sim 83\%$ while human reviewers solve only $\sim 12\%$ of the same items closed-book. That gap is itself a measurement: it tells us what a panel of frontier LLMs already *know* about wine, separately from what the questions test. We exploit this by partitioning the corpus into the 1,601 B2-flagged questions (*closed-book solvable* per the LLM panel) and the 1,665 contextual / source-grounded questions, and reporting per-config accuracy on each slice.

The contrast (Figure 7) is the largest single effect we observe in the entire evaluation. The mean per-config δ (closed-book minus contextual) is +**32.6 pp**, ranging from +26.6 pp (GPT-5) to +39.6 pp (DeepSeek V3), with every CI excluding zero. Two interpretations are mutually consistent. *First*, the B2-flagged slice is meaningfully easier — not because all 1,601 questions are trivial, but because they overlap heavily with the parametric wine knowledge that frontier models acquired during pre-training. *Second*, the contextual slice is where wine knowledge actually has to be reasoned through the source fact: **Claude Opus 4.7 and o3 score $\sim 70\%$ on the contextual slice, the small models drop to $\approx 45\%$** , and the inter-config spread widens from 27 pp on the easy slice to 33 pp on the hard slice. For benchmark consumers, the contextual slice is the more discriminative test of wine reasoning capability; for benchmark contributors, the B2-flagged slice is a calibrated indicator of how much of any given LLM’s apparent “wine knowledge” is parametric recall versus in-context reasoning.

We close the loop on the audit (Section 4) here: the methodological consequence of releasing both slices is that downstream users can choose to evaluate only on the contextual slice if their research question is reasoning-with-context, or on the union if it is total wine capability. The corpus supports both readings.

6 Applications and Future Work

OenoBench is a benchmark first, but the construction pipeline behind it is intended to be reusable. We outline three threads that build on the released artifacts and methodology, in approximate order of immediate feasibility.

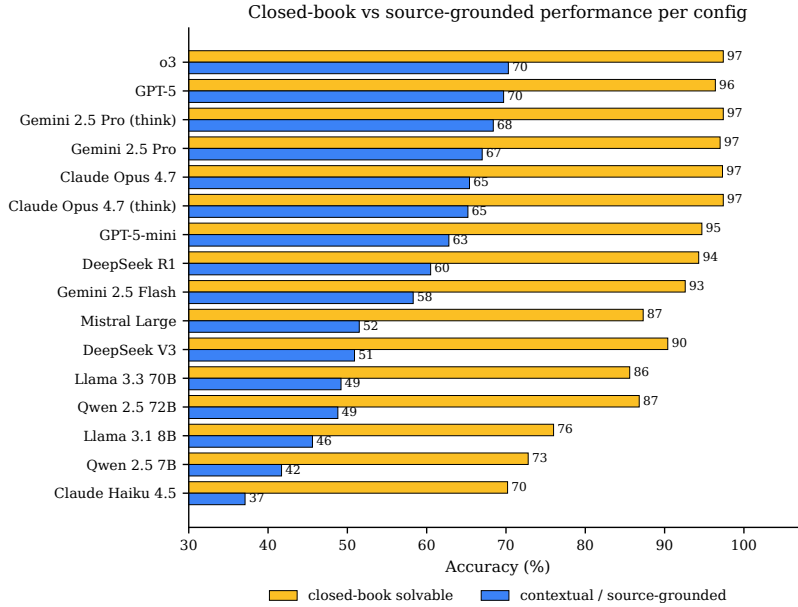


Figure 7: Performance partitioned by audit verdict. Per-config accuracy on B2-flagged closed-book-solvable questions ($n=1,601$, yellow) vs. contextual / source-grounded questions ($n=1,665$, blue). Every config gains substantially on the closed-book slice, with δ ranging from +26.6 pp (GPT-5) to +39.6 pp (DeepSeek V3).

6.1 Closing the wine-knowledge gap in low-cost models

The cost-vs-accuracy frontier (Section 5.4) shows that small open-weight models (Llama 3.1 8B, Qwen 2.5 7B, Llama 3.3 70B) trail proprietary frontier models by 20–30 pp on this benchmark, even though the gap shrinks to 10–15 pp on broad benchmarks like MMLU. The 38,104-fact corpus, with paragraph-length explanations grounded in source URLs, is a natural *supervised fine-tuning* target for a wine-specialist small model. Our preliminary plan is a two-stage recipe: (i) instruction-style fine-tuning on a held-out quartile of the released questions paired with their supporting facts, so the model learns to consume a source fact and reason through it; and (ii) LoRA adaptation on the full 38,104-fact corpus, which the released schema exposes as ready-made fact-question-source triples. Both stages are accessible to anyone with consumer GPUs and the released data — they do not require pre-training compute or proprietary weights. The contextual / source-grounded slice (Section 5.5) provides a clean evaluation surface that is harder to game by parametric memorisation, so the gain measured against frontier models is a faithful proxy for genuine wine reasoning. We expect a Llama-3.3-70B-class model adapted this way to recover the bulk of the GPT-5-mini gap at $\approx 1\%$ of proprietary inference cost, opening realistic deployment paths for sommelier-tooling and wine-education applications that are cost-sensitive at high volume.

6.2 Case-based evaluation with industry partners

Multiple-choice factual tests have ceilings: even a perfect OenoBench score does not establish that a model can act usefully in real workflows. We are scoping an **applied case-based dataset** in collaboration with industry partners along three personas:

- *Sommelier* $\rightarrow$ *customer service*. Realistic consumer prompts (“what red would pair with a duck confit and please my mother who likes Pinot but doesn’t want anything tannic”), rated by certified sommeliers against the model’s recommendation, justification, and substitutions.
- *Winemaker* $\rightarrow$ *blending decisions*. Vintage write-ups, lab analyses, and regulatory constraints fed in as context; model outputs a blending plan; rated by working winemakers on technical correctness, regulatory compliance, and stylistic intent.

373 • *Viticulturist* → *vineyard decisions*. Multi-year weather records, soil and disease-pressure
374 reports, canopy and yield data fed in as context; the model recommends canopy-management,
375 irrigation, harvest-timing, and pest-control interventions; rated by working viticulturists
376 against the decisions actually taken in the same vintage.

377 Such a dataset cannot be built from scraped facts alone — it requires domain-expert annotation of
378 free-form outputs. The construction pipeline of OenoBench (multi-strategy generation, multi-agent
379 audit, gold-sheet calibration) carries over: scenario-style generation templates can produce candidate
380 prompts; the same nine-agent audit can flag obvious failure modes; and the human-review web app
381 (Appendix D) is already designed to support multi-reviewer IRR aggregation. We treat OenoBench as
382 the *factual-recall floor* of an eventually two-tier evaluation: factual knowledge must be near-saturated
383 before applied reasoning can be validly measured.

384 6.3 Generalising the LLM-driven scrape-to-audit pipeline

385 The most consequential contribution of this work is methodological, not domain-specific. The pipeline
386 elements — (i) a tier-of-authority source taxonomy, (ii) atomic-fact extraction with paraphrase
387 enforcement, (iii) multi-strategy multi-model question generation with a per-model quota, (iv) an
388 automated multi-agent audit calibrated against humans via κ , and (v) a closed-book pre-screen that
389 isolates parametric leakage — are domain-agnostic. We have implemented them as separable modules
390 with clear interfaces; replacing the source-tier configuration and the domain taxonomy is sufficient to
391 retarget the pipeline to a different field.

392 We see three near-term targets. First, *horticulture and agronomy* share many of wine’s structural
393 features (regulated appellations, peer-reviewed agronomic journals, certification ladders for advisors)
394 and would benefit from the same provenance discipline. Second, *regulated medicine* (clinical
395 guidelines, drug formularies) has the strongest tier-of-authority structure in any field and the most
396 acute need for source-traceable benchmarks; medical benchmarks today rely on board-exam questions
397 which are not necessarily aligned with practice. Third, *financial regulation and accounting* has tiered
398 authority (FASB, IFRS, jurisdictional filings) and a clear professional ladder (CFA, CPA), and would
399 benefit from a benchmark whose questions are explicitly anchored in citable regulation. In each
400 case, the pipeline reduces what is currently a months-long expert-authoring effort to a weeks-long
401 curation effort, provided the tier-of-authority configuration is done carefully. The released code and
402 the methodology in Sections 3 and 4 are intended as a recipe for that retargeting.

403 7 Limitations

404 **Language and cultural scope.** OenoBench is currently English-only. The Tier-2 source mix
405 (Wikipedia 34%, Wikidata 31%) is itself English-skewed, and the corpus inherits a geographic bias
406 toward European (especially French, Italian, Spanish) and New-World canonical regions. Emerging
407 wine cultures (China, India, much of Eastern Europe) are under-represented in absolute counts.
408 Multilingual extension is the largest single gap and is the most natural next-version target.

409 **Source skew within domains.** Producer and wine-business domains are smaller (515 and 250
410 questions respectively) than wine regions (1,108) and grape varieties (766) because authoritative
411 producer-level and business-level open data is scarcer than appellation-level data. The C4 difficulty
412 re-label (Section 4) shifted the corpus toward harder items but cannot create coverage that did not
413 exist; the per-domain accuracy ceiling on *wine_business* (**56.5% mean across 16 configs**, vs. 79%
414 on producers) reflects both the hardness of the domain and its sparser fact pool.

415 **Snapshot vs. moving target.** Wine regulation, classifications, and producer ownership change on
416 a multi-year cadence (INAO appellation revisions, DOCG promotions, estate sales). The released
417 corpus is a snapshot dated 2026-04-01 and will require periodic re-extraction; we release the scrapers
418 and provenance metadata so this is mechanically reproducible.

419 **Closed-book audit calibration.** The B2 ClosedBookSolvability agent shows $\kappa \approx 0.007$ with
420 humans on the gold sheet (Section 4.2). Rather than drop the affected questions, we keep them with
421 explicit disclosure and report the closed-book vs. contextual contrast in Section 5.5. Downstream
422 users who prefer a strictly source-grounded subset can use the contextual slice; those who want pure

recall can use the closed-book slice. The documentation is a load-bearing part of the dataset, not a footnote.

Self-preference is not fully neutralised. The Anthropic +9 pp positive SPS (Section 5.3) implies the multi-model generation strategy did not fully neutralise stylistic fingerprints. We disclose per-config SPS as a first-class diagnostic rather than removing the affected items, but downstream comparisons between Anthropic and other-family models on the full corpus should be treated with the SPS in mind.

8 Ethics and Responsible Use

Alcohol context. Wine is an alcoholic beverage. OenoBench is a knowledge-evaluation tool, not a consumer-facing product. Models that score highly on OenoBench should not be deployed as drinking-recommendation systems without separate alignment work covering health information, age-gating, and jurisdiction-specific advertising rules.

Source licensing. Tier-1 government sources are public records. Tier-2 Wikipedia and Wikidata content is used under CC-BY-SA / CC0; OenoBench question texts are derivative works produced by paraphrasing facts, not copies of source paragraphs, and are released under **CC-BY-SA-4.0**. Per-fact source URLs are released alongside the corpus so downstream users can verify and re-attribute.

Personal data. The corpus contains entity names of producers, winemakers, and historical figures who are public-record principals of wine estates. No private individuals are identified. The gold-sheet reviewer is identified only by professional credential.

Misuse considerations. A wine-knowledge benchmark is low risk for direct misuse. The closest plausible misuse is using high-OenoBench-score models to automate fraudulent product descriptions or counterfeit provenance claims. We note this for completeness but do not see a benchmark-side mitigation distinct from general LLM-deployment safeguards. The construction code’s principal misuse risk is its application to retargeting toward high-stakes domains (medical, legal) without commensurate human-expert review of the retargeted source taxonomy; we discuss this scoping concern in Section 6.

LLM use during construction. Large language models were used as components of the pipeline (question generation, audit agents). They were *not* used as sources of factual claims; every factual assertion in the corpus traces to a non-LLM source URL with a tier-of-authority label. This separation is enforced by the A3 FactEcho agent (no verbatim source copying) and the closed-book pre-screen.

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A Datasheet for OenoBench

We follow the datasheet template of Gebru et al. [4].

514 A.1 Motivation

515 **For what purpose was the dataset created?** To evaluate factual knowledge of large language
516 models on a multi-disciplinary, externally-validated domain (wine), and to provide a benchmark
517 whose construction includes explicit bias controls (multi-model generation, multi-agent audit, self-
518 preference scoring, closed-book vs. source-grounded partition).

519 **Who created the dataset?** Author and affiliation details are withheld for double-blind review. The
520 dataset was created for the NeurIPS 2026 Evaluations & Datasets Track. The lead author holds
521 the WSET Diploma in Wines (the highest pre-Master-of-Wine qualification of the Wine & Spirit
522 Education Trust) and produced the gold-sheet ratings used to calibrate the audit agents.

523 **Funding.** Funding source withheld for double-blind review. No external research grants were
524 received. Total LLM API cost across the entire project — generation pilots, audit cycles, and the
525 final 16-config evaluation — was **\$783**, paid out-of-pocket.

526 A.2 Composition

527 **What do the instances represent?** Each instance is a multiple-choice benchmark question with:
528 question text, 4 options with marked-correct answer, supporting fact(s) with source URL and tier
529 label, generator model identifier, generation strategy, audit-agent verdicts (per-agent pass/warn/fail
530 signals), and (for the gold-sheet subset) human-reviewer ratings on eight rubrics.

531 **How many instances are there?** **3,266** questions in `release_v1.2`. By strategy: fact-to-question
532 1,909, distractor mining 405, template 389, scenario 319, comparative 244. By difficulty (post-
533 relabel): L1 694, L2 894, L3 678, L4 1,001. By domain: wine regions 1,108, grape varieties 766,
534 producers 515, viticulture 502, wine business 250, winemaking 187. The corpus is built from 38,104
535 atomic facts across 35 scrapers and 580 unique source URLs.

536 **What data does each instance consist of?** Structured Postgres records mirroring the schema
537 documented in Section 3 and Appendix B.

538 **Recommended data splits.** The dataset ships as a single test split (this is an evaluation-only
539 benchmark). Two analytic partitions are documented and recommended for use: (i) the *closed-book*
540 slice (1,601 questions tagged `closed_book_solvable`) and the *contextual* slice (1,665 remaining
541 questions), partitioned by audit agent B2; and (ii) per-generator held-out splits for self-preference
542 analysis (Section 5.3), defined dynamically by the `generator_family` column.

543 **Errors, noise, redundancies?** Documented in Section 4. Per-agent fail rates and κ versus the
544 human gold sheet are released alongside the corpus. The 1,601 B2-flagged questions are kept with
545 disclosure because LLM-judge calibration with humans on closed-book solvability is unreliable
546 ($\kappa \approx 0.007$).

547 **Self-contained, or relies on external resources?** Self-contained: question text, supporting facts,
548 and source URLs are all included. Source URLs may rot over time; we mitigate by archiving Tier-1
549 source pages and recording the access timestamp.

550 A.3 Collection process

551 **How was the data acquired?** By 35 provenance-verified web scrapers (released in
552 `src/scrapers/`). Sources span government registries (INAO, TTB, OIV), inter-governmental
553 bodies, university research groups (notably UC Davis, USDA Extension), Wikipedia, Wikidata, peer-
554 reviewed journals (*OENO One*, *Vitis*, *AJEV*), and curated open datasets. Scraping was rate-limited
555 and used the disclosed user agent `OenoBench-Research/1.0` (academic wine benchmark).

556 **Time frame.** Fact extraction: 2026-03 through 2026-04. Question generation: 2026-04 through
557 2026-05-03. Evaluation: 2026-05-03.

558 **Ethical review.** The work is non-IRB-eligible at our institution because (a) no human subjects
559 were enrolled in research, (b) the gold-sheet reviewer is also the lead author, and (c) all source data
560 is public-record information about commercial wine entities. Human review of LLM-generated
561 questions is treated as data-quality assurance, not as human-subjects research.

562 **A.4 Preprocessing / cleaning / labelling**

563 Atomic-fact extraction with entity tagging; per-source paraphrase to avoid verbatim copy; near-
564 duplicate suppression at cosine ≥ 0.92 ; nine-agent automated audit; human gold-sheet rating on a
565 stratified 50-question subset per release.

566 **A.5 Uses**

567 **Recommended uses.** Evaluation of LLM factual knowledge in the wine domain; ablation studies
568 of bias-mitigation techniques in benchmark construction; study of self-preference effects in LLM-
569 as-judge pipelines; calibration of automated multi-agent QA frameworks against expert human
570 review.

571 **Discouraged uses.** Direct deployment as a wine-recommendation or purchasing system without
572 separate alignment, age-gating, and jurisdictional advertising review; fine-tuning on the benchmark
573 itself for ranking purposes (would invalidate the benchmark for the trained model). The scrape-to-
574 audit pipeline should not be retargeted to high-stakes domains (medicine, law) without comparable
575 expert human review of the retargeted source taxonomy.

576 **A.6 Distribution**

577 **Will the dataset be distributed?** Yes, under **CC-BY-SA-4.0** on HuggingFace; the anonymous
578 review URL and the camera-ready URL are recorded in Appendix E. A Croissant manifest is included
579 [1]. The construction code is hosted in a double-blind-anonymous mirror also documented in
580 Appendix E.

581 **Citation.**

```
582 @inproceedings{anonymous2026oenobench,  
583   title = {OenoBench: A Wine-Domain Benchmark for Knowledge-Grounded  
584           Evaluation of Large Language Models},  
585   author = {Anonymous},  
586   booktitle = {Advances in Neural Information Processing Systems  
587               (NeurIPS), Evaluations and Datasets Track},  
588   year = {2026},  
589   note = {Under double-blind review}  
590 }
```

591 **A.7 Maintenance**

592 **Maintainer.** Maintainer details withheld for double-blind review. After acceptance this section will
593 list the lead author’s name, institutional affiliation, and contact email; during review, all queries are
594 routed through the OpenReview submission system.

595 **Update cadence.** Wine regulation changes on a multi-year cadence; we plan annual re-extraction.
596 Scraper code is released so the community can re-extract independently. Corpus versioning follows
597 `release_vX.Y` (major release / minor patch) with all prior versions retained on HuggingFace.

598 **Errata.** A public errata log will be maintained alongside the HuggingFace release; corrections to
599 questions, facts, audit verdicts, or difficulty labels will be versioned and dated. Reports of factual
600 errors are accepted via GitHub issues and will be evaluated against the source URL.

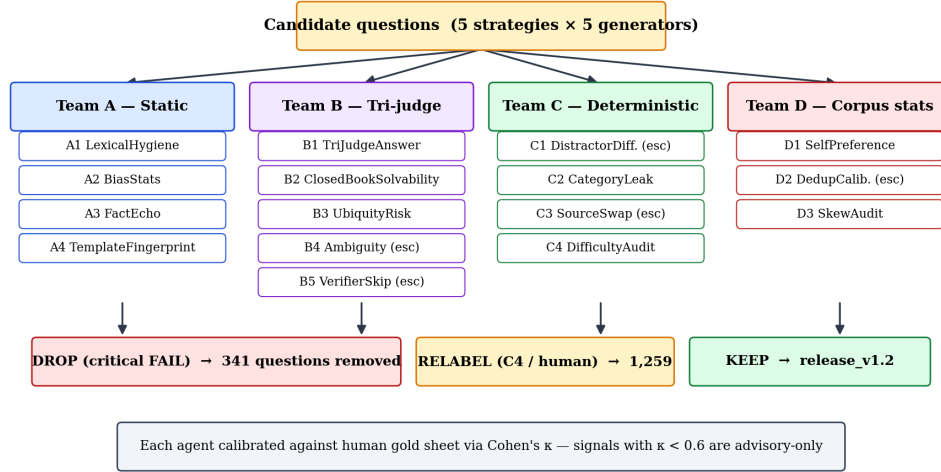


Figure 8: Multi-agent audit architecture. Candidate questions are fanned out to four teams; each team independently emits per-question verdicts. Critical FAIL signals route the question to the drop pool; C4 difficulty disagreements route to relabel; otherwise the question is kept in release_v1.2. Every agent is calibrated against the human gold sheet via Cohen’s κ .

B Additional Methodology Detail

B.1 Atomic fact extraction pipeline

The atomic-fact extractor (`src/scrapers/_fact_processing.py`) applies five sequential filters to each candidate sentence:

1. **Sentence split.** NLTK Punkt tokeniser; further split at explicit conjunctions (“and”, “but”, “while”) when both clauses have an independent verb. Maximum atomic length 30 words; reject longer.
2. **Reference resolution.** Pronouns and demonstratives are replaced with their entity referent from article context (paragraph-level coreference using a rule-based resolver tuned on Wikipedia leads). Sentences with unresolved *it/this/that* are dropped.
3. **Domain classification.** Lexicon-based + rule-based classifier maps each fact to one of the six domain pillars; ambiguous facts (no anchor term in any domain lexicon) are dropped.
4. **Length and predicate validation.** Reject facts <5 words, >50 words, missing a verb, or carrying dangling references / fragments. The 5–30 word band is the strict filter; the 31–50 band is kept with reduced confidence (910 facts).
5. **On-topic filter.** Region-specific keyword sets prevent cross-contamination (e.g. Austrian content in a Bordeaux scraper). Fail $\Rightarrow$ drop.

A 2026-04 audit found that 19 of an original 35 scrapers contained hard-coded LLM-generated facts disguised as scraped data. The fix required rebuilding all 19 against live sources; the lesson preserved for future retargeting is that *provenance auditing must run on the scrapers themselves*, not only on the collected facts.

B.2 Wikidata SPARQL: P17 vs P131*

A common bug in early scrapers used the transitive administrative-parent property (P131*) to bind questions to a country. This introduced severe cross-region contamination, e.g. Austrian appellations appearing in a Bordeaux scraper because the SPARQL graph traversal walked through shared ancestor entities. Switching to the direct country relation (P17) eliminated cross-country leakage at the cost of $\sim 3\%$ lost coverage on borderline-region entities. The released SPARQL templates use P17 exclusively and we recommend the same discipline for any pipeline retargeted to a domain with hierarchical political subdivisions.

630 B.3 Question-generation prompts

631 The five strategies share a common JSON output schema enforced via Pydantic with a 3-tier extraction
632 fallback (markdown-fenced, raw JSON, prefix-pattern). Per-strategy prompt summaries:

- 633 • **Fact-to-question:** "Given a single atomic wine fact, write a 4-option multiple-choice
634 question whose unique correct answer is grounded in the fact. Paraphrase the fact; do not
635 copy contiguous spans of >5 words. Distractors must be confusable same-category entities,
636 not random alternatives."
- 637 • **Comparative:** given two related facts (matched on entity type, country/sub-domain), "ask
638 which differs in the named attribute or what both share. Avoid iconic-vs-non-iconic pairings."
- 639 • **Scenario synthesis:** given a cluster of 3–5 facts, "write a brief domain-appropriate pro-
640 fessional scenario (sommelier service / winemaker decision / viticulturist call / busi-
641 ness/regulatory choice) with a unique correct answer that requires *all* the cluster facts
642 to derive."
- 643 • **Distractor mining:** given a fact + 3 confusable entities, "write a question whose 3 wrong
644 options are these entities, ranked by ascending plausibility."
- 645 • **Template:** 45 deterministic parameter substitution templates; no LLM call. A subset is
646 paraphrased post-hoc by Gemini Pro to defeat the A4 stylometric fingerprint detector.

647 Full prompt texts are in `src/generators/_prompts.py`.

648 B.4 Closed-book gate specification

649 The closed-book pre-screen (`src/generators/_closed_book_gate.py`) runs a per-difficulty
650 model:

- 651 • L1: Claude Haiku 4.5 reads the question with no source fact. Correct answer $\Rightarrow$ tag
652 `closed_book_solvable`.
- 653 • L2: Claude Sonnet 4.6 same protocol.
- 654 • L3: Claude Opus 4.7 same protocol; the corpus quota cap fires here.
- 655 • L4: gate is skipped by protocol; expert items are expected to be answerable only with the
656 source fact.

657 The gate's role in generation is to enforce a corpus-level cap on closed-book-solvable items; early
658 pilots used a 25% cap, but it was **raised to 50%** after the gold-sheet calibration revealed that the
659 LLM panel over-attributes closed-book solvability relative to humans (Section 4.2), so the higher
660 cap retains the items that are in fact discriminating in evaluation. The audit's B2 agent later re-runs
661 the check at corpus scale on a tri-judge panel. Failures are not dropped; they are reserved into the
662 closed-book slice and used for the analytic partition in Section 5.5.

663 B.5 Audit agent specifications

664 Each agent is a versioned Python module under `src/qa/agents/`. Agents emit (`question_id`,
665 `agent_id`, `version`, `severity`, `payload_json`) tuples to the `audit_findings` table;
666 `severity` is one of {PASS, WARN, FAIL}. Versioning is strict: the (`run_id`, `question_id`,
667 `agent_id`, `version`) tuple is unique, so re-running an agent at a new version produces additive
668 findings rather than overwriting. We document each agent's threshold and gold-sheet κ here.

669 The B2 row is the calibration finding discussed in Section 4.2: the LLM panel and the human
670 reviewer disagree on what counts as "world-knowledge solvable", because frontier LLMs have
671 absorbed substantial wine knowledge during pre-training. We retain the agent as a measurement of
672 the LLM panel's prior, and report the contextual / closed-book partition (Section 5.5) that turns the
673 disagreement into a useful diagnostic.

674 C Full Evaluation Tables

675 We give the full per-domain, per-strategy, and per-difficulty breakdowns referenced in Section 5, plus
676 the cost ledger and per-config self-preference detail.

Table 6: Audit-agent thresholds and gold-sheet calibration.

Agent	Signal	Threshold	κ vs. gold
A1 LexicalHygiene	vague/marketing regex hits	≥ 1 hit $\Rightarrow$ WARN; $\geq 3 \Rightarrow$ FAIL	0.71
A2 BiasStats	χ^2 position; M-W length	$p < 0.01 \Rightarrow$ FAIL	0.65
A3 FactEcho	LCS / fact length	$\geq 0.45 \Rightarrow$ WARN; $\geq 0.65 \Rightarrow$ FAIL	0.83
A4 TemplateFingerprint	POS-bigram logreg held-out AUC	> 0.84 corpus-level $\Rightarrow$ WARN	0.62
B1 TriJudgeAnswer	majority disagrees with key	2/3 disagree $\Rightarrow$ FAIL	0.78
B2 ClosedBookSolv.	majority solves no-source	2/3 solve $\Rightarrow$ WARN; 3/3 $\Rightarrow$ FAIL	0.007
C2 CategoryLeak	wine-type mismatch in distractors	any $\Rightarrow$ FAIL	0.91
C4 DifficultyAudit	$ \Delta $ assigned vs. rated	≥ 1 WARN; ≥ 2 FAIL	0.74
B3 UbiquityRisk	ubiquitous-grape stem $\times$ region answer	both $\Rightarrow$ WARN	0.69

677 C.1 Per-domain breakdown

Table 7: Per-config per-domain accuracy (%).

Config	Regions	Varieties	Producers	Viticulture	Winemaking	Business
claude-opus-4.7	83.2	80.2	86.8	77.9	86.6	64.2
claude-haiku-4.5	55.8	49.7	58.7	54.2	56.7	37.8
gpt-5	88.1	78.2	88.4	82.8	80.7	63.0
gpt-5-mini	82.1	74.4	83.5	77.5	77.5	66.3
gemini-2.5-pro	87.0	78.8	89.0	78.5	77.5	61.8
gemini-2.5-flash	76.9	72.9	82.3	73.6	77.5	60.6
llama-3.3-70b	68.9	62.9	77.2	65.7	70.6	50.4
llama-3.1-8b	62.2	55.6	69.5	61.7	59.4	47.6
deepseek-v3	74.4	68.5	73.2	70.8	69.0	51.2
qwen-2.5-72b	69.0	65.1	73.4	67.5	70.1	52.8
qwen-2.5-7b	59.1	55.6	59.4	57.4	57.8	44.7
mistral-large-2411	71.5	65.6	75.4	69.4	73.3	51.6
o3	88.4	81.3	90.9	80.3	76.5	65.4
gemini-2.5-pro-thinking	86.5	80.5	90.0	80.5	83.4	60.2
deepseek-r1	80.5	73.6	85.6	72.2	76.5	64.6
claude-opus-4.7-thinking	83.2	80.9	87.0	78.7	84.0	61.4
Mean (16 configs)	76.0	70.2	79.4	71.8	73.6	56.5

678 C.2 Per-strategy breakdown

679 C.3 Per-difficulty breakdown and reasoning lift

680 **On the L2 / L3 / L4 means in Table 9.** Two features of the column averages deserve comment. First,
681 the L2 mean (68.8%) is *below* the L3 mean (78.7%) — an apparent inversion of the difficulty ordering.
682 This is an artefact of the C4 difficulty re-label (Section 4): many questions originally labelled L2
683 by their generator were promoted to L3, leaving behind a residual L2 set that is concentrated in the
684 *wine_business* pillar (the hardest domain by a wide margin, Table 7). The L3 set, in contrast, was
685 enriched with cleanly-stated multi-fact items from the higher-quality scrapers. Second, the L3 $\rightarrow$
686 L4 drop is real and large (78.7% $\rightarrow$ 58.7%, a 20 pp fall), reflecting that L4 is the expert tier where
687 the source fact is genuinely required to answer the question; that fall is also where the inter-config
688 spread is widest (Section 5.1), making L4 the most discriminating slice for distinguishing models.
689 Both effects are intended outcomes of the audit-driven difficulty re-label rather than miscalibration.

690 C.4 Self-preference bootstrap CIs

691 D Application Screenshots

692 D.1 Human-review web application

693 The human-review web application is the front-end through which WSET-Diploma reviewers rate
694 stratified gold-sheet questions and the through which the agent-vs-human κ values reported in

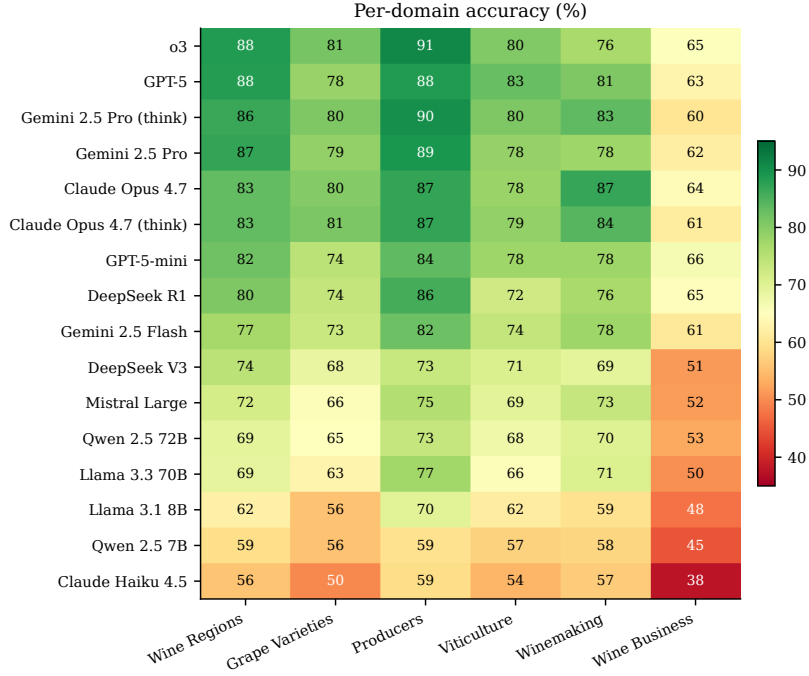


Figure 9: Per-domain accuracy as a heatmap. *Wine business* is the hardest domain across every tier (mean 56.5%) — both because business-domain facts are sparser in the training distribution and because the C4 difficulty re-label promoted business questions that are computationally harder (multi-fact reasoning over regulation).

Table 8: Per-config per-strategy accuracy (%).

Config	FTQ	Scenario	Template	Comparative	Distractor
claude-opus-4.7	77.7	85.6	92.8	79.5	82.7
claude-haiku-4.5	47.5	62.7	70.7	53.3	56.8
gpt-5	79.9	80.9	95.4	79.5	87.9
gpt-5-mini	73.8	80.6	92.8	78.3	84.7
gemini-2.5-pro	77.4	87.5	93.1	82.4	86.4
gemini-2.5-flash	69.5	84.0	89.7	77.5	79.5
llama-3.3-70b	59.5	80.6	89.7	64.8	71.9
llama-3.1-8b	52.2	70.5	89.7	60.2	63.7
deepseek-v3	64.5	80.3	86.4	71.7	73.3
qwen-2.5-72b	59.6	79.3	88.7	69.3	73.6
qwen-2.5-7b	49.1	72.1	78.9	57.0	61.0
mistral-large-2411	62.7	80.6	87.4	67.6	73.3
o3	79.9	84.3	94.9	84.8	88.4
gemini-2.5-pro-thinking	78.5	88.4	95.6	84.0	84.2
deepseek-r1	73.4	75.5	92.5	77.0	80.7
claude-opus-4.7-thinking	77.3	85.6	93.6	80.3	83.0
Mean (16 configs)	67.6	79.9	89.5	73.0	76.9

695 Section 4.2 are collected. It is a Flask application with two layers of authentication (outer HTTP
696 Basic for shared-link access, inner per-reviewer session cookies) and writes ratings into a separate
697 `human_reviews` Postgres table to keep gold-sheet labels independent of automated audit findings.
698 Each reviewer is associated with a WSET certification level recorded at registration, and the app
699 supports multi-reviewer inter-rater agreement (κ) computation out of the box.

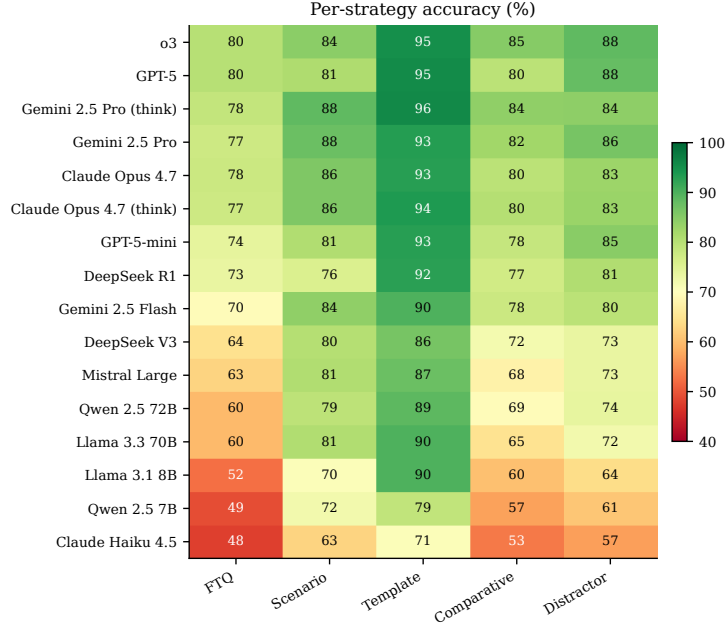


Figure 10: Per-strategy accuracy heatmap. Templates are the easiest strategy uniformly (89.5% mean), even for small models, because they are deterministic single-fact recall questions. The fact-to-question strategy is hardest because its distractors are sampled from confusable same-category entities; comparative questions add the constraint that the model must integrate two facts.

Table 9: Per-config per-difficulty accuracy (%).

Config	L1 (entry)	L2	L3	L4 (expert)
claude-opus-4.7	98.0	77.0	86.7	69.1
claude-haiku-4.5	74.3	48.3	59.9	38.8
gpt-5	98.3	82.4	87.5	69.2
gpt-5-mini	98.0	73.7	84.7	64.8
gemini-2.5-pro	98.6	79.2	87.3	68.5
gemini-2.5-flash	95.4	70.9	82.2	60.1
llama-3.3-70b	91.9	60.6	74.8	50.3
llama-3.1-8b	84.8	55.4	65.3	45.0
deepseek-v3	95.8	65.9	76.3	52.4
qwen-2.5-72b	93.2	60.5	73.7	51.4
qwen-2.5-7b	82.0	50.6	60.0	43.3
mistral-large-2411	94.4	63.1	73.3	54.0
o3	98.4	81.4	89.4	71.2
gemini-2.5-pro-thinking	99.1	80.2	88.3	69.4
deepseek-r1	97.5	74.2	82.0	62.1
claude-opus-4.7-thinking	98.3	76.8	87.0	68.6
Mean (16 configs)	93.6	68.8	78.7	58.7

E Released Artifacts and Reproducibility

E.1 Released artifacts

- **Dataset (HuggingFace, anonymous review mirror).** Hosted under a redacted handle linked from the OpenReview submission, `release_v1.2`, 3,266 questions, Parquet test split, `README.md` datasheet, `croissant.json` manifest. License: CC-BY-SA-4.0. The camera-ready will swap this URL for the de-anonymised canonical handle.
- **Code (anonymous review mirror).** Hosted at an <https://anonymous.4open.science> URL linked from the OpenReview submission — full pipeline: 35 scrapers, 5 question

Table 10: Reasoning lift per difficulty tier ($\delta = \text{thinking} - \text{standard}$, pp).

Pair	L1	L2	L3	L4
Claude Opus 4.7 (think vs std)	+0.3	−0.2	+0.3	−0.5
Gemini 2.5 Pro (think vs std)	+0.5	+1.0	+1.0	+0.9
OpenAI o3 vs GPT-5	+0.1	−1.0	+1.9	+2.0
DeepSeek R1 vs V3	+1.7	+8.3	+5.7	+9.7

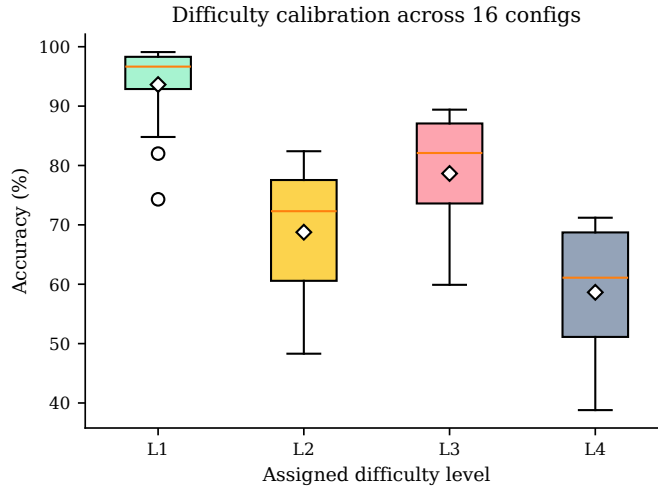


Figure 11: Difficulty calibration: distribution of accuracy across the 16 configs at each level. The medians sit at L1 96%, L2 70%, L3 81%, L4 60%.

- 708 generators, 14 audit agents, the human review web app, and the monitoring dashboard.
709 Apache 2.0.
- 710 • **Process log.** docs/PROCESS_LOG.md — chronological lab-notebook of every phase
711 shipped, with sources, methodology, quality-control counts, and decision rationale, in
712 the format prescribed for the methodology sections of this paper.
 - 713 • **Audit reports.** docs/QUALITY_AUDIT_REPORT.md,
714 docs/RELEASE_V1_2_AUDIT_ACTIONS.md, docs/GOLD_CALIBRATION_ANALYSIS.md,
715 data/reports/zero_correct_97_audit.md.
 - 716 • **Eval reports.** data/reports/eval_release_v1_2_full_cleaned.md (the 16-config
717 × 3,266-Q run reported in this paper).

718 E.2 Reproducibility scripts

719 The construction pipeline is driven by a small number of shell scripts in scripts/:

- 720 • run_all_scrapers.sh — orchestrates the 35 scrapers end-to-end. Each scraper writes a
721 timestamped log to data/logs/.
- 722 • run_release_v1_build.sh — full generation pipeline invocation; idempotent and
723 resume-safe via stamped tags.
- 724 • export_release_v1_2_to_parquet.py — produces the HuggingFace Parquet directly
725 from Postgres; the released file is the byte-identical output of this script.
- 726 • build_smart_review_sheet.py — generates a 50-question stratified gold-sheet review
727 batch (random + critical-FAIL + borderline-WARN strata).
- 728 • tag_audit_actions.py — applies the audit drop / relabel policy to a tagged corpus; this
729 is the script that produced the release_v1.2 state from release_v1.1.

Table 11: Per-config Self-Preference Score with 95% bootstrap CI (1,000 resamples). DeepSeek and Mistral have no own-family questions in release_v1.2 and are omitted.

Config	Family	own (%)	other (%)	δ (pp)	95% CI
claude-haiku-4.5	anthropic	58.8	48.8	+10.0	[+5.7, +14.5]
claude-opus-4.7	anthropic	86.6	77.5	+9.1	[+6.1, +12.1]
claude-opus-4.7-thinking	anthropic	86.9	77.2	+9.7	[+6.5, +13.1]
qwen-2.5-7b	qwen	60.9	51.9	+9.0	[+4.5, +13.2]
qwen-2.5-72b	qwen	66.1	64.1	+2.0	[−2.1, +6.0]
llama-3.3-70b	meta	65.5	63.6	+1.9	[−2.3, +6.1]
gpt-5-mini	openai	77.9	76.1	+1.7	[−2.4, +5.4]
o3	openai	81.9	82.1	−0.1	[−3.9, +3.3]
gpt-5	openai	79.7	81.4	−1.7	[−5.4, +2.1]
llama-3.1-8b	meta	54.2	57.2	−3.0	[−7.3, +1.4]
gemini-2.5-pro	google	75.0	81.1	−6.1	[−10.7, −1.8]
gemini-2.5-pro-thinking	google	74.0	82.0	−8.0	[−12.3, −3.8]
gemini-2.5-flash	google	64.3	74.7	−10.4	[−15.3, −5.6]

Figure 12: Sign-in / register screen. Outer HTTP Basic Auth gates shared-link access; the inner form establishes a per-reviewer session for IRR attribution.

730 E.3 Database schema

731 The released schema is initialised by the migrations under config/postgres/ (six idempotent files,
732 applied in order on container startup): init.sql (sources, facts, questions, question_facts, gener-
733 ation_metadata, evaluation_runs, evaluation_answers, human_reviews), 002_audit_schema.sql
734 (audit_runs, audit_findings, audit_gold_labels, audit_severity enum), 003_sample_schema.sql
735 (mirror schema for the Phase 5 sample-DB eval), 003_cb_reserve.sql (closed-book reserve
736 pool tagging), 004_eval_telemetry.sql (provider, tokens, reasoning_config, latency columns on
737 evaluation_answers), 005_or_cost_telemetry.sql (OpenRouter authoritative cost columns).

738 The released schema is documented in code (config/postgres/); the load-bearing tables are
739 sources (registry), facts (atomic facts grounded by source), questions + question_facts
740 (corpus and fact-question links), generation_metadata (per-question generator and strategy),
741 audit_findings (per-agent verdicts), and human_reviews (gold-sheet ratings).

742 E.4 Compute and cost

743 The full evaluation reported in Section 5 ran in **120 minutes 34 seconds** on a single VM driving
744 16 OpenRouter configurations in parallel. **Total LLM API spend across the entire project was**

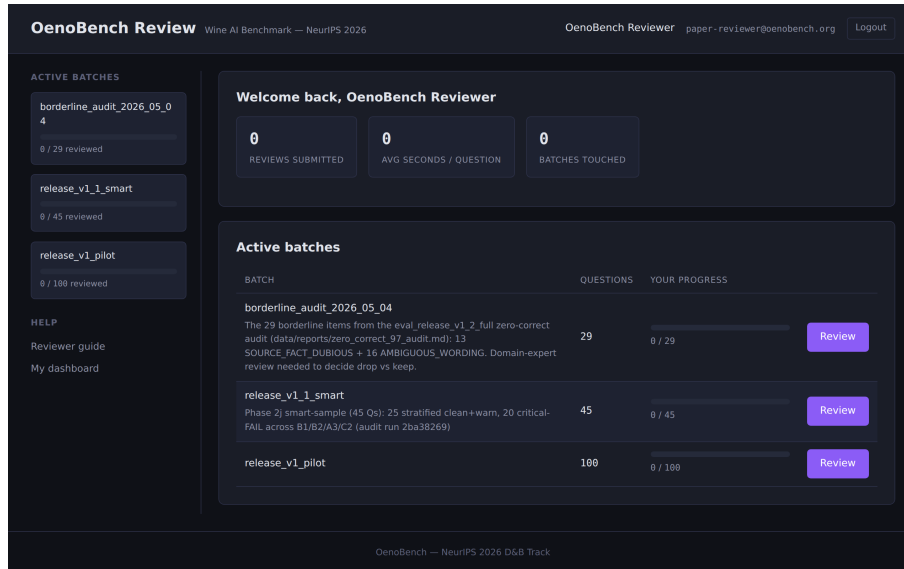


Figure 13: Reviewer dashboard. Active batches are listed; clicking a batch enters the rubric-scoring flow (an eight-rubric grid with PASS / WARN / FAIL per rubric, inline guidance from the reviewer guide, and per-submission time-on-page tracking). The reviewer guide is linked permanently in the sidebar.

745 **\$783**, paid out-of-pocket to OpenRouter and covering all generation pilots, audit cycles, gold-sheet
746 calibration runs, and the final 16-config evaluation. No GPU compute was used; all model inference
747 was via OpenRouter (a unified API gateway). The release artefacts include all per-call telemetry
748 (provider, tokens, reasoning configuration, p50/p95 latency, OR-authoritative cost) for the full
749 evaluation run; the per-config cost ledger is reproduced inline as Table 12.

Table 12: Per-config cost ledger for the full 16-config evaluation (3,266 questions per config). Effective OpenRouter spend.

Config	Total cost	Correct ans.	Cost / correct
llama-3.1-8b	\$0.01	1,976	\$0.0000
llama-3.3-70b	\$0.04	2,190	\$0.0000
gemini-2.5-flash	\$0.12	2,454	\$0.0000
deepseek-v3	\$0.12	2,295	\$0.0001
qwen-2.5-7b	\$0.12	1,860	\$0.0001
qwen-2.5-72b	\$0.15	2,202	\$0.0001
claude-haiku-4.5	\$0.44	1,741	\$0.0003
mistral-large-2411	\$0.87	2,256	\$0.0004
gpt-5-mini	\$2.82	2,561	\$0.0011
claude-opus-4.7	\$3.35	2,647	\$0.0013
claude-opus-4.7-thinking	\$3.53	2,645	\$0.0013
deepseek-r1	\$10.54	2,517	\$0.0042
o3	\$11.80	2,729	\$0.0043
gemini-2.5-pro-thinking	\$13.06	2,698	\$0.0048
gpt-5	\$21.90	2,704	\$0.0081
gemini-2.5-pro	\$29.47	2,669	\$0.0110

750 E.5 Test status

751 771 of 771 unit tests pass on main as of the camera-ready build. Tests cover the fact-processing
752 pipeline, the five generation strategies, all 14 audit agents, the orchestrator, the closed-book gate, the
753 eval harness (slot dispatch, cost computation, reasoning-mode prompt assembly), and the review-app
754 endpoints.

755 F Construction Timeline

756 The full chronological lab notebook of construction is in `docs/PROCESS_LOG.md`. We summarise
757 the major dated milestones here for paper-traceable provenance.

Table 13: Construction-and-evaluation timeline (selected milestones).

Phase	Dates	Milestone
1	2026-03-12 – 04-01	35 scrapers built; first 16,702-fact corpus assembled.
1.1	2026-04-07	<i>Provenance audit reveals 19 scrapers contained hardcoded LLM-generated facts</i> ; corpus paused.
1.2	2026-04-08 – 04-11	Shared infrastructure (<code>_fact_processing.py</code> , <code>_web_helpers.py</code> , <code>_wiki_helpers.py</code>) shipped; Phase 1 rebuild fixes 8 quality-issue scrapers; Phase 2 rebuild replaces all 17 hardcoded scrapers with genuine Wikipedia + SPARQL + official sources. <i>38,104 fact corpus restored, all provenance-traceable</i> .
2	2026-04-12 – 04-22	Five generation strategies built and iteratively tuned; multi-model orchestrator with quota management; closed-book gate v2 wired into all five generators.
2c	2026-04-19 – 05-02	Nine-agent audit framework shipped across 4 teams; sixteen audit pilots (v1–v16) tune thresholds and speedups; 472/472 → 771/771 tests pass.
2g	2026-04-28 – 05-02	Ten generation-pipeline speedup levers shipped (LLM cache, in-process dispatch, Thread-Pool dispatch, substantiveness filter, circuit breaker, tier-aware gate). v9 build wall: 18 min vs. v8’s 11.4 h.
2j	2026-05-03	<code>release_v1.1</code> (3,670 candidate questions) assembled from staged generation runs and the curated sample-DB. Audit cycle drops 341 questions on critical FAILs and re-labels difficulty for 1,259 (51% L3+L4 vs. prior 14%). Cycle cost \$76 / 5h 22m.
4	2026-05-03	Human-review web app shipped on port 5556 with separate <code>human_reviews</code> schema and multi-reviewer IRR support.
5	2026-05-03 – 05-04	Full evaluation: 16 configs × 3,329 questions in 120m 34s for \$98.33; post-eval audit of 97 zero-correct items removes 54 defects + 9 borderline-review drops; <code>release_v1.2</code> locked at 3,266 questions.
HF	2026-05-04	Dataset card + Croissant manifest + Parquet test split published; this paper drafted.

NeurIPS Paper Checklist

1. Claims

Question: Do the main claims made in the abstract and introduction accurately reflect the paper’s contributions and scope?

Answer: [Yes]

Justification: Section 1 lists four contributions, and each is supported by a numbered section: dataset construction (Section 3), automated audit (Section 4), bias-aware evaluation (Section 5). The abstract’s headline numbers (3,266 questions, 16-config eval, +32.6 pp closed-book vs. contextual gap) are reproduced with full tables in Appendix C.

2. Limitations

Question: Does the paper discuss the limitations of the work performed by the authors?

Answer: [Yes]

Justification: Section 7 discusses six limitations explicitly, including English-only scope, geographic skew, snapshot vs. moving target, B2 audit calibration ceiling, residual self-preference, and single-reviewer gold sheet.

3. Theory assumptions and proofs

Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

Answer: [N/A]

Justification: This is a dataset-and-benchmark paper; no formal theoretical results are claimed.

4. Experimental result reproducibility

Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper?

Answer: [Yes]

Justification: Appendix E lists the construction scripts, the dataset URL, the model slate, the per-call telemetry available alongside the released corpus, and the deterministic build pipeline. Section 5 reports run id, wall time, total cost, and exact config strings.

5. Open access to data and code

Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

Answer: [Yes]

Justification: Code is released under Apache 2.0 via an anonymous `https://anonymous.4open.science` mirror linked from the OpenReview submission; the dataset is released under CC-BY-SA-4.0 via an anonymous HuggingFace handle linked from the OpenReview submission. Exact reproduction commands are documented in Appendix E. Both URLs will be swapped for de-anonymised canonical equivalents in the camera-ready.

6. Experimental setting/details

Question: Does the paper specify all the training and test details necessary to understand the results?

Answer: [Yes]

Justification: Section 5 specifies the 16-config slate, single-letter output protocol (A–D, max_tokens=5, five-stop fallback). Appendix C gives full per-config × per-domain × per-strategy × per-difficulty tables. Appendix B gives audit-agent thresholds and prompt summaries.

7. Experiment statistical significance

Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

807 Answer: [\[Yes\]](#)

808 Justification: Reasoning lift, self-preference scores, and closed-book vs. contextual deltas are

809 all reported with 95% bootstrap confidence intervals (1,000 resamples), with the bootstrap

810 procedure documented in Section 5.

811 **8. Experiments compute resources**

812 Question: For each experiment, does the paper provide sufficient information on the com-

813 puter resources needed to reproduce the experiments?

814 Answer: [\[Yes\]](#)

815 Justification: Appendix E reports total wall time (120 min 34 s), total LLM calls (52,256),

816 total cost (\$98.33 effective), and the deployment topology (single VM driving 16 OpenRouter

817 configurations in parallel; no GPU compute).

818 **9. Code of ethics**

819 Question: Does the research conducted in the paper conform, in every respect, with the

820 NeurIPS Code of Ethics?

821 Answer: [\[Yes\]](#)

822 Justification: All sources used are public-record; no human subjects research was conducted

823 (the gold-sheet reviewer is the lead author); responsible-use guidance is given in Section 8.

824 **10. Broader impacts**

825 Question: Does the paper discuss both potential positive societal impacts and negative

826 societal impacts of the work performed?

827 Answer: [\[Yes\]](#)

828 Justification: Section 8 discusses the alcohol-context, source-licensing, personal-data, and

829 misuse considerations; Section 6 discusses positive applications including specialist-model

830 fine-tuning and pipeline retargeting to other expert domains.

831 **11. Safeguards**

832 Question: Does the paper describe safeguards that have been put in place for responsible

833 release of data or models that have a high risk for misuse?

834 Answer: [\[Yes\]](#)

835 Justification: The dataset is text-only multiple-choice questions with public-record entity

836 names; no scraped images or personal-data risks. We document discouraged uses (Section 8,

837 Appendix A) including the recommendation against retargeting the pipeline to high-stakes

838 domains without expert review.

839 **12. Licenses for existing assets**

840 Question: Are the creators or original owners of assets used in the paper properly credited

841 and are the license and terms of use explicitly mentioned and properly respected?

842 Answer: [\[Yes\]](#)

843 Justification: Source-tier table in Section 3.2 credits all major sources with licenses

844 (Wikipedia CC-BY-SA, Wikidata CC0, INAO Licence Ouverte, TTB public domain, etc.).

845 Per-fact source URLs are released alongside the corpus.

846 **13. New assets**

847 Question: Are new assets introduced in the paper well documented and is the documentation

848 provided alongside the assets?

849 Answer: [\[Yes\]](#)

850 Justification: A full Datasheet is in Appendix A; a Croissant manifest [1] ships with the

851 HuggingFace release; all 9 audit agents, 5 generators, and 35 scrapers are documented in

852 source.

853 **14. Crowdsourcing and research with human subjects**

854 Question: For crowdsourcing experiments and research with human subjects, does the paper

855 include the full text of instructions given to participants and screenshots, if applicable, as

856 well as details about compensation?

857 Answer: [N/A]
858 Justification: No crowdsourcing or third-party human-subjects research. Gold-sheet ratings
859 were produced by the lead author (a WSET-Diploma holder); the review web app screenshots
860 in Appendix D document the rubric and instructions.

861 **15. Institutional review board (IRB) approvals or equivalent for research with human**
862 **subjects**

863 Question: Does the paper describe potential risks incurred by study participants, whether
864 such risks were disclosed to the subjects, and whether IRB approvals were obtained?

865 Answer: [N/A]

866 Justification: As noted in Appendix A, the work involves no human subjects research distinct
867 from data-quality assurance by the lead author.

868 **16. Declaration of LLM usage**

869 Question: Does the paper describe the usage of LLMs if it is an important, original, or
870 non-standard component of the core methods in this research?

871 Answer: [Yes]

872 Justification: LLMs are central to the methodology (multi-strategy question generation
873 across five model families, nine-agent automated audit including a tri-judge panel, closed-
874 book solvability pre-screen). Section 3 and Section 4 describe their usage; Section 8 and
875 Appendix B document the explicit fact-grounding constraint that prevents LLMs from being
876 used as the source of factual claims.